

Bradley Pascoe

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Experience

Research

Research Fellow University of Southern Queensland – Springfield, QLD Jun 2025 – Present

- Redesigned research code for vehicle codesign optimisation into modular, tested Python libraries, enabling reuse across multiple studies and improving reproducibility, numerical methods, and performance.
- Developed a Python library for numerical computation of vehicle radiation spectra, accelerating core algorithms with Cython for high-performance evaluation.

Doctoral Candidate, Research Associate University of Sydney – Sydney, NSW Mar 2021 – Mar 2025

- Investigated the effects of strain on turbulent mixing layers by implementing and validating advanced numerical methods into a Fortran codebase for the solution of partial differential equations.
- Utilised high-performance computing infrastructure using code developed for OpenMP and MPI parallelisations to simulate and analyse terabytes of data, aided by Bash and Python scripts to organise jobs and files.
- Automated visualisations of simulated data with Python APIs, reducing time and memory usage by 20-75%.
- Derived and calibrated reduced order models, achieving accurate predictions of fluid flow quantities for over $10,000\times$ less computational cost compared to simulating the three-dimensional fluid flow.

Research Engineer Defence Science and Technology Group – Melbourne, VIC Feb 2020 – Jan 2021

- Enhanced the pre- and post-processing capabilities of multi-dimensional fatigue spectrum and strain data by developing and optimising a library in R, including introducing version control and unit tests.
- Designed programs to read and evaluate data from fatigue testing systems, allowing for continuous verification of experimental regulatory compliance.

Teaching

Academic Tutor/Demonstrator

Engineering Mechanics (AMME1802), University of Sydney	2024
Engineering Methods (AMME3060), University of Sydney	2024
Applied Fluid Dynamics and Turbulence (AMME5292), University of Sydney	2024
Linear Systems and Control (MMAN3200), University of NSW	2018, 2019, 2021, 2022
Engineering Mechanics (ENGG1300), University of NSW	2018, 2019
Engineering Mechanics 2 (MMAN2300), University of NSW	2018, 2019
Mechanics of Solids (MMAN2400), University of NSW	2018, 2019
Physics 1A (PHYS1121), University of NSW	2018, 2019

Mentoring

Supervised an engineering undergraduate thesis project (AMME4112) 2023

Education

University of Sydney – PhD in Engineering	Dec 2024
University of New South Wales – B.Eng.(Hons) in Aerospace Engineering	Jan 2021
University of New South Wales – B.Sc. in Physics	Jan 2021

Skills

Languages: Bash, C++, Fortran, MATLAB, Python, R, SQL

Tools: Git, LaTeX, MPI, OpenMP

Software: ANSYS Fluent, CATIA, Paraview, SolidWorks, Tecplot

Publications

- **B. Pascoe**, M. Groom, D.L. Youngs, and B. Thornber (2026), Impact of transverse strain on linear, transitional and self-similar turbulent mixing layers, *Journal of Fluid Mechanics*, 1030, A61
- **B. Pascoe**, M. Groom, B. Thornber (2026), Late-time growth of an inhomogeneous, turbulent mixing layer subjected to transient compression, *Physica D: Nonlinear Phenomena*, 488, 135056
- **B. Pascoe**, M. Groom, B. Thornber (2025), Turbulence modelling of mixing layer under anisotropic strain, *Physical Review Fluids*, 10, 064609
- **B. Pascoe**, M. Groom, D.L. Youngs, and B. Thornber (2024), Impact of axial strain on linear, transitional and self-similar turbulent mixing layers, *Journal of Fluid Mechanics*, 999, A5

Presentations

- *Shock-driven compressible turbulent mixing in planar and convergence geometries*, University of Oxford (2024)
- *The impact of strain-rates on linear, transitional and self-similar turbulent mixing layers*, First Light Fusion (2024)
- *Reynolds-averaged Navier-Stokes modelling for the Richtmyer-Meshkov instability under anisotropic strain*, 18th International Workshop on the Physics of Compressible Turbulent Mixing (2024)
- *Modelling the effects of convergence on the Richtmyer-Meshkov instability in planar geometry*, American Physical Society Division of Fluid Dynamics Annual Meeting (2023)
- *Modelling the Richtmyer-Meshkov instability under strain*, International High Performance Computing Summer School (2023)
- *The Richtmyer-Meshkov instability under axial strain*, 23rd Australasian Fluid Mechanics Conference (2022)
- *The effect of axial strain on the Richtmyer-Meshkov instability*, 17th International Workshop on the Physics of Compressible Turbulent Mixing (2022)

Grants

- Very High-Order Solver Frameworks for Compressible Turbulent Mixing, UK Research and Innovation GPU eCSE, £127,992 (2025)

Honours and Awards

- University of Sydney HDR Publication Award - Mechanical/Thermofluids (2024)
- International High Performance Computing Summer School attendee (2023)
- Science Vacation Research Scholarship (2018)
- Future Physicist International Summer Camp attendee (2018)

Service

Journal/Conference Reviewer

Journal of Fluid Mechanics, *Physica D: Nonlinear Phenomena*, 24th Australasian Fluid Mechanics Conference

Elected Student Representative Council Member

Australasian Fluid Mechanics Society

2023 – 2024

- Chair of the HDR/ECR subcommittee and the Emerging Leaders subcommittee.

Conference Administration Lead

University of Sydney

Sep 2022 – Jan 2023

- Prepared the delegate materials, childcare services, and catering, and managed a team of 30 volunteers for a four-day conference with 400+ delegates, helping the conference to achieve a rating of 4.57/5 for organisation.
- Designed and distributed the post-conference survey and compiled the results into a report, obtaining a record number of responses for the post-conference survey.